5A - 6A - 10A - Atmosphere - EvolLand

# Test 3 Master Test Review

*38 Review Questions*

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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**REVIEW**

# Test 3 Master Test Review

*38 questions worth focusing on as you prepare for Test 3, organized by subject. Each one tells you where to find the answer. Work through them in your own words, typing your answers on the lines.*

| WHERE TO FIND THE ANSWERS | PASSWORD |
|---|---|
| Module 5A | Module5A |
| 5AE1 | 5AE1SG |
| 5AE2 | 5AE2SG |
| Atmospheric Chart | Atm |
| Atmospheric Pressure exercise | AtmosSG |
| Ancient Atmospheres video | AASG |
| EvolLand exercise | EvolSG |
| Module 6A | Module6A |
| 6AE1 | 6AE1SG |
| Module 10A | Module10A |

## The Earth's Atmosphere

**1.** What is the atmosphere? Excluding water vapor, name the four main gases that make it up, with their percentages.

**Find answers in:** Module 5A, 5AE1

**2.** Name the four layers of the atmosphere in order, from the ground up, and the "pause" at the top of each. What is a pause?

**Find answers in:** Module 5A, 5AE1, Atmospheric Chart

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**3.** In which two layers does the temperature DECREASE with increasing altitude, and in which two does it INCREASE? What heats each of the two warming layers?

**Find answers in:** Module 5A, 5AE1, Atmospheric Chart

**4.** In which layer is the ozone layer, and what does it do? Which two gases in that layer protect life from deadly UV radiation? Why is ozone “good up high, bad nearby”?

**Find answers in:** Module 5A, 5AE1

**5.** What is the ionosphere, and where is it? Up to what frequency does it reflect radio waves?

**Find answers in:** Module 5A, 5AE1

**6.** What sphere harmlessly traps the charged particles of the solar wind?

**Find answers in:** Module 5A, 5AE1

**7.** What is Earth-based weather? Name the six main ingredients of Earth-based weather.

**Find answers in:** Module 5A, 5AE2

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**8.** Name the four forms of precipitation, and describe each.

**Find answers in:** Module 5A, 5AE2

**9.** Are fog, dew, and clouds forms of precipitation? Explain.

**Find answers in:** Module 5A, 5AE2

**10.** In summer, what are cumulonimbus (storm) clouds primarily composed of?

**Find answers in:** Module 5A, 5AE2

**11.** What is a greenhouse gas? Name three. Why are nitrogen and oxygen NOT greenhouse gases?

**Find answers in:** Module 5A, 5AE2

**12.** What makes high and low pressure, and what weather does each bring?

**Find answers in:** Module 5A, 5AE2 (Homework #11)

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## Atmospheric Pressure Exercise

**13.** Define pressure and atmospheric pressure. How does atmospheric pressure change as you go higher, and why?

**Find answers in:** Atmospheric Pressure exercise

**14.** Give the value of one Atmosphere in millibars, psi, and inches of mercury (in Hg).

**Find answers in:** Atmospheric Pressure exercise

## Ancient Atmospheres

**15.** What was the Great Oxygenation Event, and which organisms caused it?

**Find answers in:** Ancient Atmospheres video (Module 5A, Homework #6)

## EvolLand Exercise

**16.** About how many millions of years ago was the Cambrian Explosion, and what happened in it?

**Find answers in:** EvolLand exercise

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**17.** As oxygen in the atmosphere rose, life moved from the sea to the land. About how high did oxygen peak, and when? What happened to flying insects?

**Find answers in:** EvolLand exercise (Module 5A, Homework #6)

**18.** In years, about how long ago did our species, Homo sapiens, evolve?

**Find answers in:** EvolLand exercise

## The Nature of Light

**19.** What is optics?

**Find answers in:** Module 6A, 6AE1

**20.** What is the simple definition of light?

**Find answers in:** Module 6A, 6AE1

**21.** Rounded, what is the speed of light in a vacuum, in km/s and in miles/s?

**Find answers in:** Module 6A, 6AE1

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**22.** Who first used astronomy to determine the speed of light, when, and how?

**Find answers in:** Module 6A

**23.** What does it mean to say laser light is coherent?

**Find answers in:** Module 6A, 6AE1

**24.** List the electromagnetic spectrum in order of increasing frequency. About what range of wavelengths can humans see? Are cosmic rays part of the spectrum?

**Find answers in:** Module 6A, 6AE1

**25.** What are the three ways light interacts with matter? Define each.

**Find answers in:** Module 6A, 6AE1

**26.** Name and define the five classical phenomena of light. Which model of light explains all five, and which one can the particle model adequately explain?

**Find answers in:** Module 6A, 6AE1

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**27.** What is the difference between specular and diffuse reflection? Do all shiny objects exhibit specular reflection?

**Find answers in:** Module 6A

## The Exploration of Space

**28.** What was Sputnik 1? Who launched it?

**Find answers in:** Module 10A

**29.** What was the first satellite the United States successfully launched? What did it discover?

**Find answers in:** Module 10A

**30.** Who was the first person to orbit the Earth?

**Find answers in:** Module 10A

**31.** Who were the first and second Americans in space?

**Find answers in:** Module 10A

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**32.** Who was the first American to orbit the Earth?

**Find answers in:** Module 10A

**33.** How many Apollo missions set out to land on the Moon, how many succeeded, and how many men have walked on the Moon?

**Find answers in:** Module 10A

**34.** What famous “first” in aviation belongs to Charles Lindbergh? In what plane, and from where to where?

**Find answers in:** Module 10A

**35.** Which uncrewed mission produced the first close-up images of Jupiter? What else was first about it?

**Find answers in:** Module 10A

**36.** Which uncrewed mission produced the first close-up images of Saturn?

**Find answers in:** Module 10A

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**37.** What were the Viking missions, and what was their key first?

**Find answers in:** Module 10A

**38.** Which spacecraft is the only one to have visited Uranus and Neptune?

**Find answers in:** Module 10A

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